



HOTEL MANAGEMENT SYSTEM

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Introduction

The assignment will involve developing the Hotel Management System by systematically building its various components, implementing essential features, and ensuring secure and efficient functionality. The process will include creating the admin login system, managing rooms, customers, and bookings, and implementing the room reservation functionality. Test cases will be conducted to ensure the system operates correctly under various scenarios.

This document will cover the detailed implementation of each feature, the testing procedure used, and a UML diagram representation of the codebase, to provide a comprehensive understanding of the system's design and functionality.

Benefits of the Hotel Management System

- **Simplified Operations:**
Automates and simplifies key administrative tasks, such as customer registration, check-ins, and check-outs, reducing manual effort and improving efficiency.
- **Centralized Management:**
Provides a one place to manage all aspects of hotel administration, including rooms, customers, bookings, and staff, allowing for easier oversight and control.
- **Enhanced Accuracy:**
Minimizes the risk of human error by automating data entry and calculations, ensuring that room availability, booking details, and financial transactions are accurate.
- **Improved Customer Experience:**
Facilitates a smoother check-in and check-out process, ensuring a better experience for guests through efficient management of their stay.
- **Real-Time Data Access:**
Allows administrators to access up-to-date information on room availability, customer bookings, and financial reports, enabling prompt decision-making.
- **Comprehensive Reporting:**
Offers detailed reports through Crystal Reports, providing valuable insights into various aspects of hotel operations for better strategic planning.
- **Enhanced Security:**
Implements secure login and authentication processes to protect sensitive data and ensure that only authorized personnel can access and manage the system.
- **Scalability:**
Designed to accommodate future growth and additional features, allowing the system to evolve alongside the needs of the hotel.

T 01. Analysis and Modelling of the Hotel Management System Requirements

Functional Requirements

- **Login Authentication**
 - The system must allow administrators to log in using a username and password.
 - The system must validate credentials and grant access only to authorized users.
- **Dashboard Interface**
 - The system must provide a dashboard with quick access to customer registration, check-in, and check-out functionalities.
 - The dashboard must display a photo collage of the hotel and its name.
- **Customer Registration**
 - The system must allow administrators to enter and save new customer information in the database.
 - The system must validate customer details before saving.
- **Customer Check-In**
 - The system must allow administrators to book an available room for a customer on a preferred date.
 - The system must update room availability and customer records accordingly.
- **Customer Check-Out**
 - The system must allow administrators to check out customers, set the check-out date to today, and calculate the total charge based on the number of days stayed.
 - The system must update room availability and billing records.
- **Room Management**
 - The system must allow administrators to add, update, or remove room details from the database.
 - The system must manage room types, rates, and availability.
- **Customer Management**
 - The system must allow administrators to edit, delete, or add customer details.
 - The system must maintain accurate customer records.
- **Booking Management**
 - The system must allow administrators to view and remove booking records.
 - The system must update the database to reflect changes in bookings.
- **Admin Management**
 - The system must allow administrators to view, add, or delete admin accounts and manage their credentials.

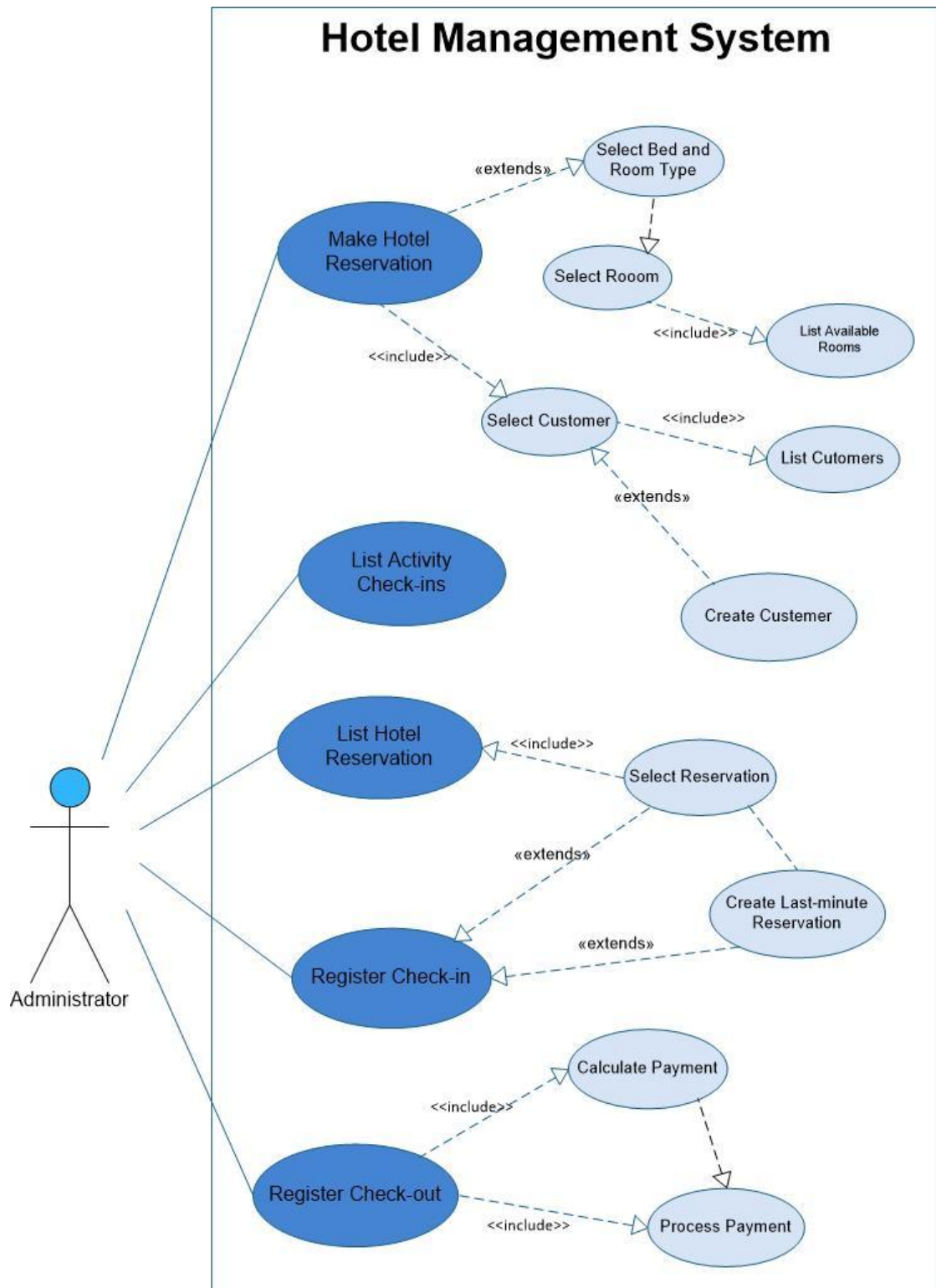
- **Report Generation**
 - The system must generate Crystal Reports for Rooms, Customers, Bookings, and Admins tables.
 - The system must provide customizable reporting options.
- **Logout Functionality**
 - The system must allow administrators to securely log out of their session.

Non-Functional Requirements

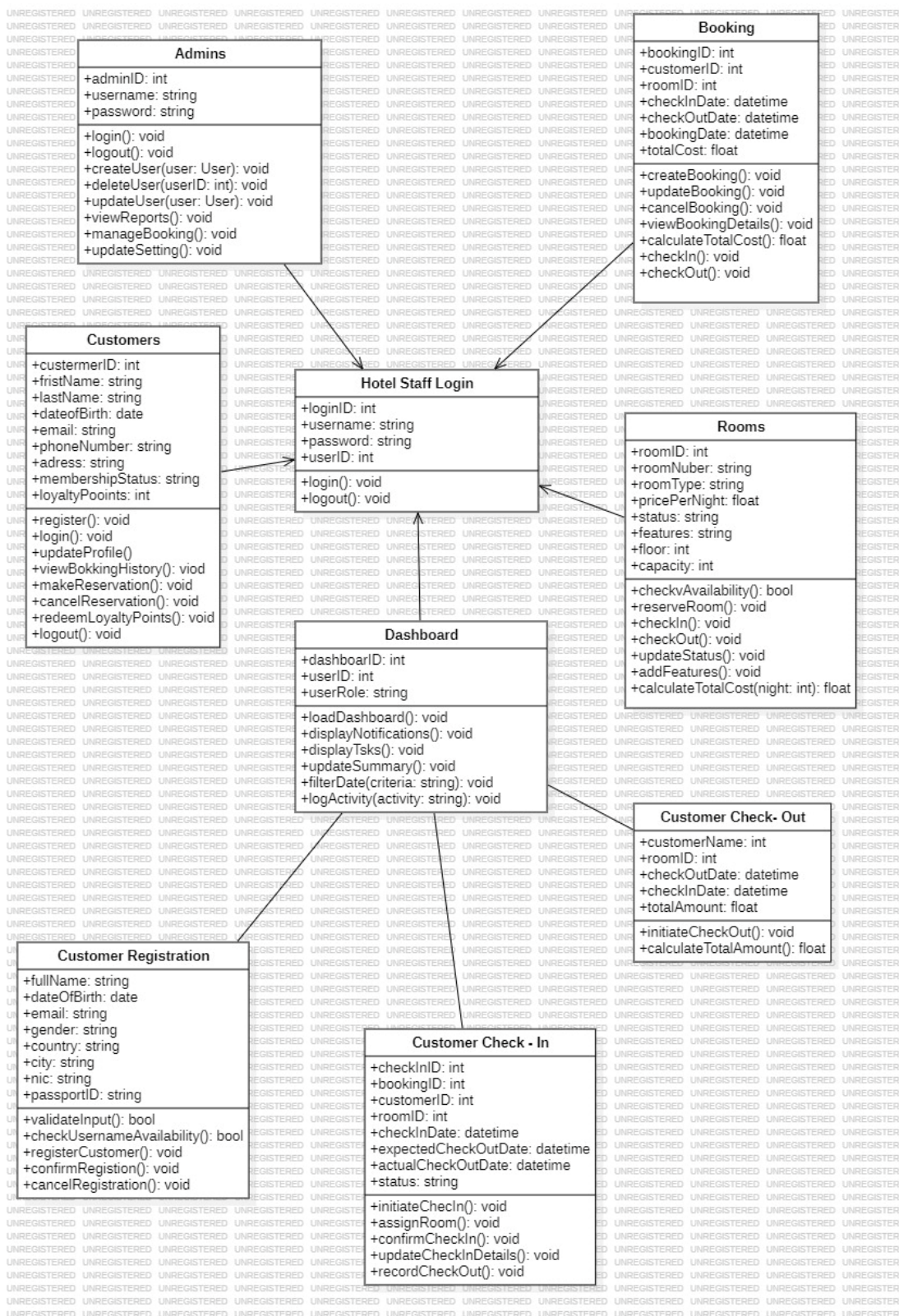
- **Performance**
The system should respond to user actions within a reasonable time frame, ensuring a smooth user experience.
- **Usability**
The system should have an intuitive and user-friendly interface, allowing administrators to navigate and perform tasks efficiently.
- **Security**
The system must protect sensitive data through encryption and secure login mechanisms. It must ensure that only authorized users can access and modify data.
- **Reliability**
The system should be stable and operate without unexpected crashes or data loss. It must handle errors gracefully and provide meaningful error messages.
- **Maintainability**
The system's codebase should be well-documented and organized to facilitate future updates and maintenance.
- **Compatibility**
The system should be compatible with common operating systems and web browsers used by administrators.
- **Backup and Recovery**
The system should support regular data backups and provide mechanisms for data recovery in case of system failure.
- **Accessibility**
The system should be accessible to users with varying levels of technical expertise and provide adequate help and support.
- **Compliance**
The system must comply with relevant data protection regulations and standards.

UML Diagrams

Use case Diagram

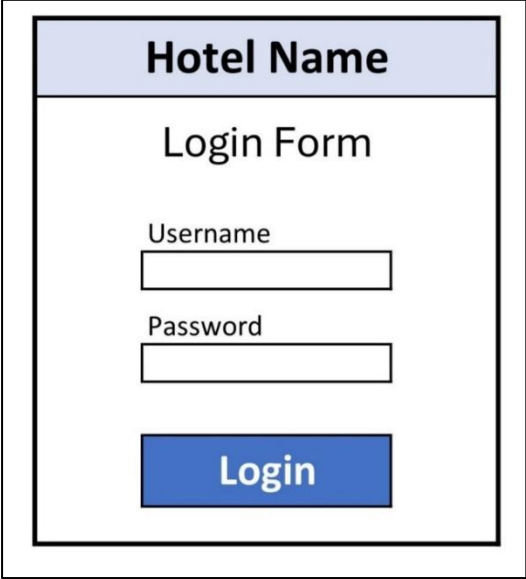


Class Diagram



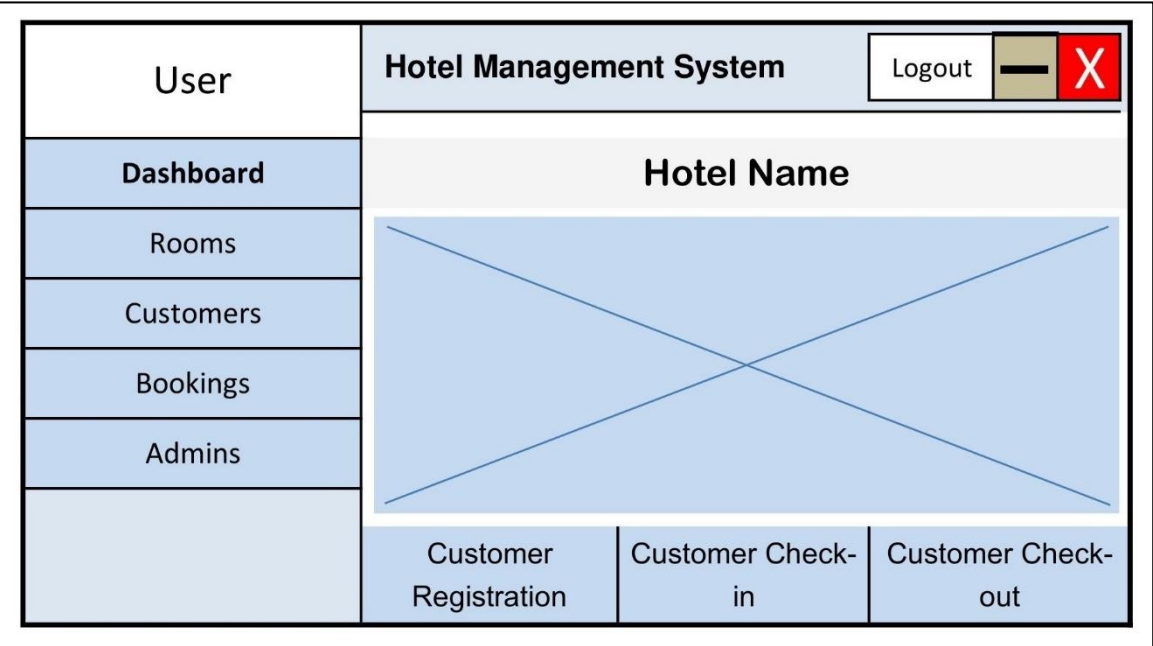
User Interface (UI) Sketches

To ensure an intuitive and user-friendly interface for the Hotel Management System, detailed UI sketches are created as a foundational step in the design process. These sketches will outline the layout and functionality of each interface. By visualizing the user interface in advance, we can identify and address potential usability issues, streamline the user experience, and ensure that the final design is both visually appealing and easy to navigate.



A sketch of an admin login form. It features a light blue header bar with the text "Hotel Name". Below the header is a white box containing the title "Login Form". Inside this box, there are two input fields: "Username" and "Password", each with a corresponding text label above it. At the bottom of the white box is a blue button with the text "Login".

Figure 1.1 - Admin Login Form1



A sketch of a dashboard interface. It has a light blue header bar with the text "Hotel Management System" on the left, a "Logout" button in the center, and a red "X" button on the right. Below the header is a white box containing the title "Hotel Name". The main content area is a large blue rectangle with a large "X" drawn across it. At the bottom of the dashboard, there are three blue buttons: "Customer Registration", "Customer Check-in", and "Customer Check-out".

Figure 1.2 - Dashboard

	Customer Registration		Logout	—	X
Dashboard	Customer Name	City			
Rooms	Phone No.	Country			
Customers	Email Address	Gender			
Bookings	Date of Birth	NIC			
Admins	Register				

Figure 1.3 - Customer Registration

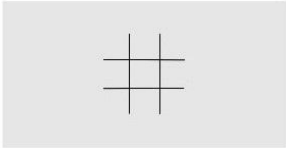
	Customer Check-In		Logout	—	X
Dashboard	Search	New Customer Registration			
Rooms			Check-In Date		
Customers			Bed Type		
Bookings			Room Type		
Admins			Room No.		
			Price		
	Customer ID		Check-In		
	Customer Name				
	NIC				
	Clear				

Figure 1.4 - Customer Check-In

	Customer Check-Out		Logout	—	X
Dashboard	Search Customer				
Rooms					
Customers					
Bookings					
Admins					
	Customer Name	Check-In Date			
	Room No.	Check-Out Date	Price		
	Check-Out		Clear		

Figure 1.5 - Customer Check-Out

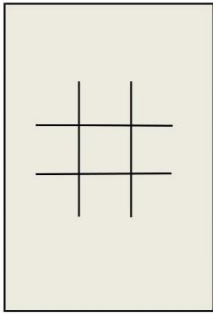
Rooms		Logout	—	X
Dashboard Rooms Customers Bookings Admins		Room No		
		Room Type		▼
		Bed		▼
		Price		
		Status		▼
		Add Update		
Delete Clear				

Figure 1.6 - Rooms

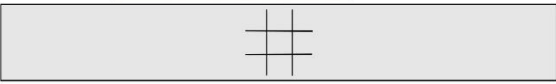
Customers		Logout	—	X	
Dashboard Rooms Customers Bookings Admins	Search				
					
	Customer Name		City		Add
	Phone No.		Country		Update
	Email Address		Gender		Delete
	Date fo Birth		NIC		Clear

Figure 1.7 - Customers

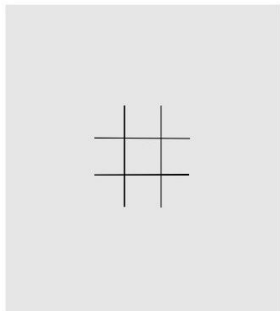
Bookings		Logout	—	X
Dashboard Rooms Customers Bookings Admins		Booking ID		
		Customer ID		
		Room No		
		Delete		
		Clear		

Figure 1.8 - Bookings

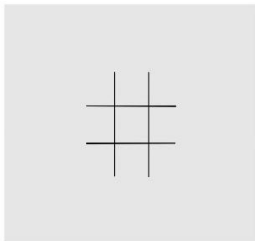
Admins		Logout	—	X
Dashboard Rooms Customers Bookings Admins	Search			
				
	Admin ID			
	Username			
	Password			
	Add Delete			
Clear				

Figure 1.9 - Administrators

T 02. Object-Oriented Development and Database Integration

User Interfaces (GUI)

This section describes how the user will interact with the system through different window forms and passes.

The user interface for the Hotel Management System is designed to provide a seamless experience for the hotel administrators. It includes separate interfaces for each management area, ensuring ease of use and customized functionalities.

The Hotel Management System is designed as a desktop-based application with a fixed width of 1366px and a height of 768px. This specific size ensures that the application provides a consistent and optimal user experience on desktop devices.

Login Form

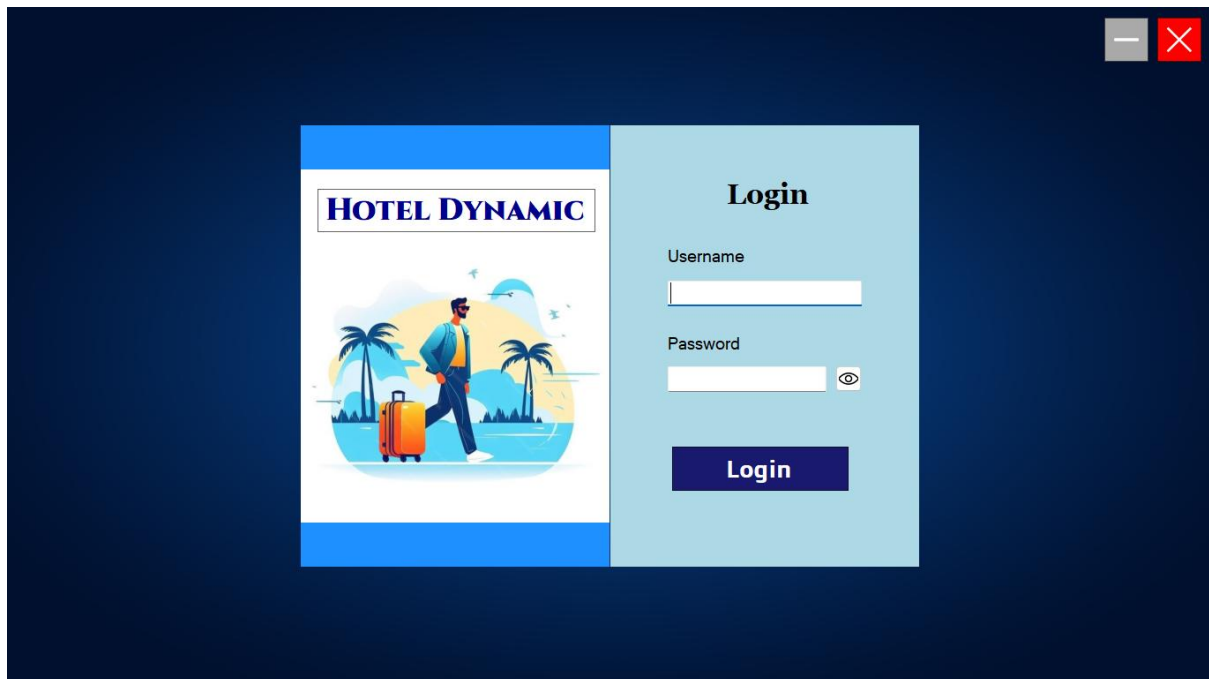


Figure 1 – Admin Login Form

The Login interface (figure 1) of the Hotel Management System is designed to provide a secure entry point for administrators. This interface offers users the option to log in using their username and password. The 'Login' button will direct the user to the system Dashboard (figure 2), when a username and the matching password is entered to the relevant fields.

[Default : **Username** – admin, **Password** - 12345]

Dashboard

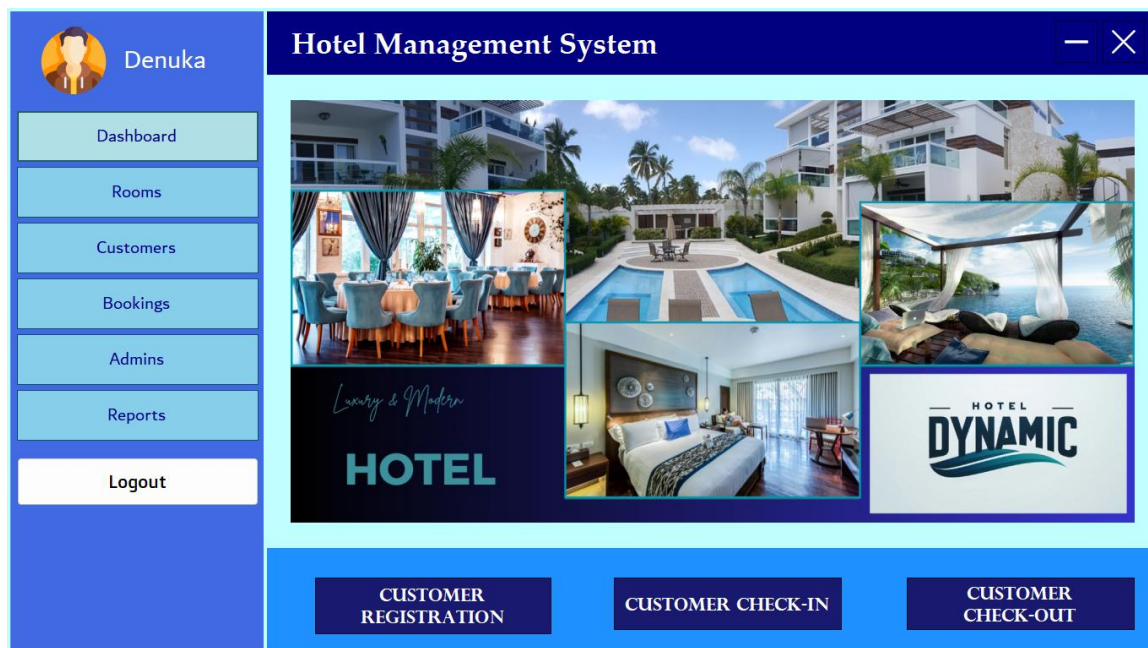


Figure 2 - Dashboard

The Dashboard features buttons for Customer Registration, Check-In, and Check-Out, a navigation menu with buttons for Dashboard, Rooms, Customers, Bookings, and Admins, and a photo collage of hotel with hotel logo. The design aims for intuitive navigation and quick access to key management functions.

Customer Registration

The "Customer Registration" form is presented within a window titled "Customer Registration". It includes the same sidebar as the dashboard. The form fields are organized into two columns: "Customer Name", "City", "Phone No.", "Country", "Email Address", "Gender" (a dropdown menu), "Date of Birth" (a date picker showing "01-01-2000"), and "NIC / Passport ID". A "Register" button is positioned at the bottom right of the form area.

Figure 3 – Customer Registration

The Customer Registration interface features navigation buttons on the left and a logout button at the bottom. The central form captures customer details (name, phone no., E-mail, birth date, city, country, gender, ID no.), with a "Register" button to finalize the process.

Customer Check-In

Customer Check-In

Search

New Customer Registration

ID	Name	NIC
1	John Smith	200215368945
2	Isabel Diaz	200156897442
4	Samman Perera	98562415786V
5	Bobbina Samson	65984123587
9	Jihan	200226001763
10	Ravi	20001781542
11	Harthaa	205136482211

Customer ID

Customer Name

NIC / Passport ID

Check-In Date

Bed

Room Type

Room Number

Price (for one Night) Rs.

CHECK-IN

Clear

Figure 4 – Customer Check-in

Registered Customers can make reservations through the Check-In interface of the Hotel Management System. Today's date or an upcoming date can be picked as the Check-in date. Available rooms with the selected Bed and Room Type will display on the Combo Box. With "Check-In" button booking can be recorded in the database.

Customer Check-Out

Customer Check-Out

Customer Name Search Customer

ID	CustomerID	CustomerName	RoomNo	CheckinDate	CheckoutDate	Price	CheckOut
1	1	John Smith	101	6/1/2024	6/3/2024	6750	Yes
2	2	Isabel Diaz	102	6/1/2024	6/7/2024	12000	Yes
4	5	Bobbina Samson	105	6/4/2024	6/6/2024	9250	Yes
6	1	John Smith	101	6/7/2024	7/7/2024	27750	Yes
7	9	Jihan	104	6/7/2024	6/8/2024	6000	Yes
8	10	Ravi	104	6/6/2024	6/8/2024	6750	Yes
9	11	Harthaa	105	6/8/2024	7/7/2024	43000	Yes
10	4	Samman Perera	101	7/7/2024	7/8/2024	6000	Yes

Customer Name

Check-In Date

Check-Out Date (today)

Room Number

Get Price

Price (Rs.)

CHECK-OUT

Clear

Figure 5 – Customer Check-out

All the bookings made through the Hotel Management System will be displayed on the Data Grid View with the 'CheckOut' status. When a record with the Checkout status "No" is selected, relevant text fields will be filled. "Get Price" button will determine the price for the days stayed and "Check-out" button will complete the process.

Rooms

RoomNo	RoomType	Bed	Price	DailyCharge	Status	Booked
101	AC	Single	6000	750	Available	No
102	Non-AC	Double	7000	1000	Available	No
103	Non-AC	Single	5000	500	Available	No
104	AC	Single	6000	750	Available	No
105	AC	Double	8000	1250	Available	No
106	Non-AC	Triple	9000	1500	Available	No
107	Non-AC	Single	5000	500	Available	No
108	Non-AC	Double	7000	1000	Available	No

Room Number

Room Type

Bed

Price (in LKR)
[for one Night]

Daily Charge per Additional Night

Status

Figure 6 - Rooms

Administrators have control over rooms in hotel through the Room interface, enabling them to add new rooms, modify existing ones, or delete rooms as needed. This interface streamlines management tasks, ensuring the hotel's accommodation offerings are always up-to-date and accurate.

Customers

Search

ID	Name	PhoneNo	Email	DateOfBirth	City	Country	Gender	NIC
1	John Smith	0785694123	jonsmith@gmail.com	3/12/2002	London	England	Male	200215368945
2	Isabel Diaz	0745689231	isdiaz@gmail.com	9/28/1999	Madrid	Spain	Female	200156897442
4	Saman Perera	0778513629	samp@gmail.com	8/25/1998	Galle	Sri Lanka	Male	98562415786V
5	Bobbina Samson	0712354682	bobsam@gmail.com	4/13/2000	Chennai	India	Female	65984123587
9	Jhan	0786916838	jhanaklanka65@	6/16/2002	galle	sri lanka	Male	200226001763
10	Ravi	0768747173	ravi@123gmail.com	1/3/2001	colombo	Sri Lanka	Male	20001781542
11	Harthaa	0766535273	harthaa@gmail.com	8/17/2001	London	England	Female	205136482211

Customer Name

City

Phone No.

Country

Email Address

Gender

Date of Birth

NIC / Passport ID

Figure 7 - Customers

The Customer interface enables administrator to manage customers in the hotel database. Customers in the database can be searched by the name, new customers can be added, existing customers can be removed or their details can be updated as necessary.

Bookings

Denuka

Dashboard

Rooms

Customers

Bookings

Admins

Reports

Logout

Bookings

ID	CustomerID	CustomerName	RoomNo	CheckinDate	CheckoutDate	Price	CheckOut
1	1	John Smith	101	6/1/2024	6/3/2024	6750	Yes
2	2	Isabel Diaz	102	6/1/2024	6/7/2024	12000	Yes
4	5	Bobina ...	105	6/4/2024	6/6/2024	9250	Yes
6	1	John Smith	101	6/7/2024	7/7/2024	27750	Yes
7	9	jhan	104	6/7/2024	6/8/2024	6000	Yes
8	10	Ravi	104	6/6/2024	6/8/2024	6750	Yes
9	11	Hanthaa	105	6/8/2024	7/7/2024	43000	Yes
10	4	Saman P...	101	7/7/2024	7/8/2024	6000	Yes

Booking ID

Customer ID

Room Number

Delete

Clear

Figure 8 - Bookings

The Booking interface provides administrators with the ability to manage hotel reservations effectively. Through this interface, administrators can delete existing bookings, ensuring accurate and up-to-date reservation records. This functionality helps maintain smooth operations and prevents overbooking or inconsistencies in the booking system.

Administrators

Denuka

Dashboard

Rooms

Customers

Bookings

Admins

Reports

Logout

Administrators

Search

ID	Username	Password	image
1	admin	12345	D:\IDM Computing\CF pro...
3	Denuka	12345678	D:\IDM Computing\CF pro...
5	jhan	2002	D:\IDM Computing\CF pro...
6	Siva	25948	D:\IDM Computing\CF pro...
7	New	1234	D:\IDM Computing\CF pro...

Admin ID

ID

Username

Password

Icon

Browse



Add

Remove

Clear

Figure 9 - Administrators

The Admin Page facilitates the addition and removal of new administrators from the database effortlessly. It offers intuitive controls and secure access, ensuring efficient management of administrative privileges within the system.

Reports

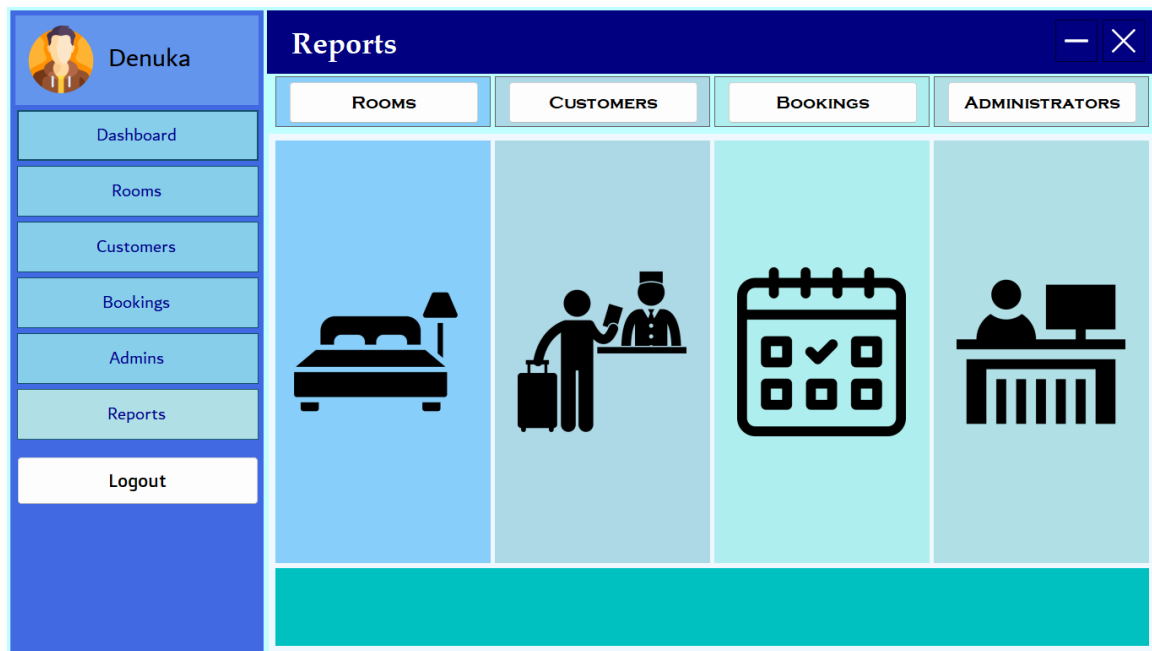


Figure 10 - Reports

Administrators can generate detailed reports through the Reports interface of the Hotel Management System. Crystal Reports for Rooms, Customers, Bookings, and Admins can be created; by selecting a button, administrators can view a comprehensive report that presents all relevant data. The system ensures that the reports are accurate and up-to-date, providing valuable insights into hotel operations for better decision-making.

Reports in the Hotel Management System

The Reports feature in the Hotel Management System provides administrators with detailed and customizable insights into various operational aspects of the hotel. Utilizing Crystal Reports, the system generates comprehensive reports on rooms, customers, bookings, and admin activities. These reports are designed to offer a clear view of performance metrics, booking trends, and customer data, enabling informed decision-making and effective management.

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Hotel Management System Database Tables

To manage customer information, room details, and bookings efficiently, ensuring smooth hotel operations and effective administrative management, a Microsoft Access database was made for the Hotel Management System.

Login Information for Admin

Field name	Data Type	Description
Id	int	Administrator Identity Number
Username	Short Text	Unique can be a name or email
Password	Short Text	A combination of letters, numbers and symbols

Customers

Field name	Data Type	Description
Id	int	Customer Identity Number
Name	varchar(200)	Full Name of the Customer
PhoneNo	varchar(50)	Contact information
Email	varchar(200)	Personal Email address
DateOfBirth	DateTime	Customer's date of birth
City	varchar(100)	City of customer's home address
Country	varchar(100)	Homeland of customer
Gender	varchar(20)	Male / Female / Other
NIC	varchar(100)	Proof of Id; NIC number or Passport Id

Rooms

Field name	Data Type	Description
RoomNo	varchar(20)	Unique room number
RoomType	varchar(100)	Air-Conditioned / non-AC
Bed	varchar(100)	Bed Size: Single / Double/ Triple
Price	varchar(50)	Price for a single night
DailyCharge	varchar(100)	Price for additional nights staying
Status	varchar(100)	Availability
Booked	varchar(50)	Current status of room: Booked or not

Bookings

Field name	Data Type	Description
Id	int	Booking Identity Number
CustomerID	Short Text / varchar(50)	Customer Identity Number
CustomerName	Short Text / varchar(200)	Name of the customer
RoomNo	Short Text / varchar(20)	Selected Room number
CheckinDate	DateTime	Check-in Date
CheckoutDate	DateTime	Check-out Date
Price	Short Text / varchar(50)	Total price for the stay
CheckOut	Short Text / varchar(20)	Checkout status: Yes/No

T 03. Testing and Implementation of the Hotel Management System

The testing process of the Hotel Management System involves designing comprehensive test cases to ensure the application functions correctly under various scenarios. Each test case includes specific input conditions, expected outcomes, and actual outcomes.

The process covers functional testing for features like room management, customer registration, and bookings, as well as security and usability testing. By systematically executing these test cases, we can identify and address any defects, ensuring the system is reliable, user-friendly, and secure.

Test Case	Function	Process	Expected Outcome	Actual Outcome	Status
T1	Admin Login	Login with a valid username and password.	Application Dashboard will display with the Success Message	Dashboard displayed with message box "Login Successful"	Success
T2	Admin Login	Login with a blank username and password.	Display Error Message, "Both fields are required"	As Expected	Success
T3	Admin Login	Login with a valid username and an incorrect password.	Display Error Message, "Incorrect Password"	Displayed error message "Incorrect username or password"	Partially Success
T4	Customer Registration	Submit the form with the "Customer Name" field left empty.	Display an error message indicating the "Customer Name" field is required	Error message asking the user to Fill all the fields is shown	Partially Success
T5	Customer Registration	Enter an invalid phone number in the "Phone No." field.	Display an error message saying "Invalid Phone Number"	Doesn't display an error message, but the text field only accepts numbers	Partially Success
T6	Customer Check-in	Click on "New Customer Registration" button	The user gets redirected to the "Customer Registration" interface	As Expected	Success
T7	Customer Check-in	"Check-In" without filling all the text fields	Display an error message requesting to Fill all the fields	As Expected	Success

Test Case	Function	Process	Expected Outcome	Actual Outcome	Status
T8	Customer Check-out	Search Customer button with the text field being empty.	Display Error Message, "Please enter a customer name"	No error message, all the records displayed on the data grid view	Fail
T9	Customer Check-out	Select a booking with Checkout status "Yes".	Text fields remain empty, details of the selected booking do not display	As Expected	Success
T10	Customer Check-out	"Get Price" without selecting a booking.	Display Error Message, "Please select a booking"	Error message "Select a check-in record" displayed	Success
T11	Rooms	"Add", "Update", "Remove" with the fields being empty.	Show error message asking the user to Fill the fields or Select a record	As Expected	Success
T12	Rooms	Select a preferred "Room Type" and a "Bed" type.	Price of a room, with selected Bed and Room Type fill the relevant fields	As Expected	Success
T13	Customers	Enter an email address in "E-mail", that doesn't exist.	Display an error message indicating the "Invalid Email address"	Does not display an error, any type of text is accepted.	Fail
T14	Customers	"Add", "Update", "Remove" with blank text fields.	Display error message asking the user to Fill the fields or Select a record	As Expected	Success
T15	Bookings	Select a booking from the DataGrid View.	Customer and Room details of the selected Record, appear on the relevant fields	As Expected	Success
T16	Bookings	"Delete" without selecting a booking.	Display Error Message, "Please select a booking"	Error message, "Select the record that needs to be deleted" is shown	Success
T17	Admins	"Add", "Remove" with empty text fields.	Show error message prompting the user to Fill the fields or Select a record	As Expected	Success
T18	Admins	Enter a text in password field and click "Show" Button.	Text in password field as '*', will display as normal, and vice versa	As Expected	Success

Conclusion

The development of the Hotel Management System offered a valuable opportunity to practice building, coding, and testing a complete administrative solution using Visual Studio, C# and MS Access. This experience helped me understand core features like customer registration, room and booking management, and generating reports. By applying object-oriented principles and integrating a well-structured database, I ensured that the system was both efficient and met the required specifications.

The UML class diagram provided a clear visualization of how different system components interact, giving a comprehensive view of the code structure and functionality.

Overall, the assignment has been an excellent opportunity to refine my skills in software design, implementation, and testing. It underscored the importance of organized code and rigorous testing to develop a reliable and user-friendly application.